

SQL DATA ANALYSIS REPORT

1. Objective & Declarative Methodology

The goal of this project is to apply relational database logic to transition raw spreadsheet records into structured business intelligence. Unlike procedural coding languages that loop through data row-by-row, this analysis utilizes **declarative SQL**, defining the exact required output and allowing the internal database engine to optimize the retrieval path.

2. Core Queries and Result Sets

2.1 High-Level Revenue & Operational Metrics

- **Business Goal:** Retrieve total transaction scale, aggregate revenue, and baseline order norms.
- **SQL Query Syntax:**

SQL

```
SELECT
  COUNT(*) AS total_orders,
  SUM(TotalPrice) AS total_revenue,
  AVG(TotalPrice) AS average_order_value
FROM orders;
```

- **Result Set Evidence:**
 - total_orders: 1,200
 - total_revenue: \$1,264,761.96
 - average_order_value: \$1,053.97

2.2 Product Performance Breakdown

- **Business Goal:** Aggregate individual transactional rows to identify top-grossing items.
- **SQL Query Syntax:**

SQL

```
SELECT
  Product,
  SUM(Quantity) AS total_units_sold,
  SUM(TotalPrice) AS total_gross_revenue
FROM orders
```

GROUP BY Product
ORDER BY total_gross_revenue DESC;

- **Result Set Evidence:**
 - **Chair:** 562 units sold | \$195,620.11 total revenue
 - **Printer:** 542 units sold | \$195,612.61 total revenue
 - **Laptop:** 535 units sold | \$192,126.56 total revenue
 - **Tablet:** 497 units sold | \$186,568.95 total revenue
 - **Monitor:** 480 units sold | \$175,651.41 total revenue
 - **Desk:** 508 units sold | \$167,459.93 total revenue
 - **Phone:** 411 units sold | \$151,722.39 total revenue

2.3 Marketing Channel Attribution

- **Business Goal:** Evaluate acquisition counts and financial generation across referral channels.
- **SQL Query Syntax:**

SQL

```
SELECT
  ReferralSource,
  COUNT(*) AS customer_acquisition_count,
  SUM(TotalPrice) AS channel_gross_revenue
FROM orders
GROUP BY ReferralSource
ORDER BY channel_gross_revenue DESC;
```

- **Result Set Evidence:**
 - **Instagram:** \$275,285.45 gross revenue
 - **Email:** \$261,808.55 gross revenue
 - **Google:** \$250,441.48 gross revenue
 - **Facebook:** \$250,410.90 gross revenue
 - **Referral:** \$226,815.58 gross revenue

3. Executive Business Recommendations

1. **Prioritize High-Margin Categories:** **Chairs** and **Printers** represent the highest absolute revenue sources for our platform. Marketing budgets should be skewed to heavily emphasize these categories rather than low-volume items like Phones.
2. **Optimize Social Media Ad Spend:** **Instagram** is the clear leader in customer acquisition revenue (\$275,285.45). This channel should be prioritized for upcoming visual ad campaigns and product placements.